

CATEGORY

USB - HARD DRIVES

	Brand	Freecom	Freecom	Freecom	Freecom	Iomega
		FHD-1 40 GB	FHD-1 80 GB	FHD-2 20 GB	FHD-2 40 GB	80 GB
FEATURES (30%)	Drive Specifications:					
	Physical Dimensions (L x W x H) (mm)	186 x 112 x 47	186 x 112 x 47	140 x 80 x 19	140 x 80 x 19	184.2 x 114.3 x 34.3
	Weight (in gms)	900	900	190	190	817
	Drive Form factor (in inches)	3.5	3.5	2.5	2.5	3.5
	Capacity (in GB)	40	80	20	40	80
	Spindle speed (in rpm)	7200	7200	4200	4200	7200
	Data Buffer size (in MB)	2	2	2	2	2
	Interface type	USB 2.0	USB 2.0	USB 2.0	USB 2.0	USB 2.0
	Visual Indicators	Power, Data transfer and Synchronise	Power, Data transfer and Synchronise	Power and Data transfer	Power and Data transfer	Power cum Data transfer
	Additional configurable buttons	1, Sync data	1, Sync data	✗	✗	✗
	Build Quality					
	Sturdiness (Scale of 5)	3	3	4	4	3
	Aesthetic design (Scale of 5)	4	4	3.5	3.5	3.5
	Package contents					
	Bundled software	Fsync 2.0	Fsync 2.0	Acronis TrueImage	Acronis TrueImage	Iomega AutoBackup, Norton Ghost 2003, Iomega Fat32 formatter tool
PERFORMANCE (70%)	Carry-case included (Y/N)	✗	✗	✗	✗	✗
	Other accessories	Power cable, data cable	Power cable, data cable	Power cable, data cable	Power cable, data cable	Power cable, data cable
	Documentation (Y/N)					
	Quickstart guide	✓	✓	✓	✓	✓
	Manual	on CD	on CD	on CD	on CD	on CD
	Sisoft Sandra Professional 2003 Drive Benchmark					
	Drive Index (in KBps)	22687	22631	14591	18904	22036
	Sequential Read (in MBps)	33	31	21	27	31
	Random Read (in MBps)	7	8	4	5	7
	Sequential Write (in MBps)	27	27	21	26	27
	Random Write (in MBps)	9	12	5	7	11
	Average Access Time (in milliseconds)	8	6	11	11	7
	HDTach 2.7					
	Random Access Time (in msecs)	16	13.8	19.3	20.9	18.8
	Average Read Speed (in MBps)	24	27.1	18.6	23.3	28.2
OVERALL SCORE	Average Write Speed (in MBps)	14.7	18.7	12.3	13.9	18.3
	Read Burst Speed (in MBps)	34.3	34.3	34.2	33.9	34.4
	CPU utilisation (%)	14.2	19.7	10	12.8	15.1
	Real world file transfer tests					
	Sequential Read (1 GB) (in secs)	51.5	65.48	106.74	47	51.04
	Assorted Read (1 GB) (in secs)	53.66	52.09	61.69	53.86	56.9
	Sequential Write (1 GB) (in secs)	45.41	115.33	55.11	84.35	59.07
	Assorted Write (1 GB) (in secs)	58.33	55.62	73.92	62.76	87.69
	Price per MB: (in Rs)	0.35	0.22	0.71	0.47	0.14
	Features	9.00	9.00	11.30	11.30	7.90
	Performance	60.66	59.40	47.00	53.42	57.60
	Overall Score (Performance + Features)	69.66	68.40	58.30	64.72	65.50
	Value for Money	9.83	15.78	4.11	6.85	23.29
	Grade	C	D	C	C	D+
	Price (Rs)	13,500	17,000	13,500	18,000	11,000

tests, the maximum drive index score of 22,687 was logged by the 7,200-rpm 40 GB Freecom FHD-1 drive, followed by its 80 GB sibling. Most of the 2.5-inch form factor drives logged below the 15,000 mark, which highlights the disparity in performance due to the difference in the spindle speeds. The Freecom FHD-2 40 GB model was hampered by its slower speeds, but still managed to record a good drive index of 18,904.

The sequential read test was a more closely fought battle, with very little variations between the scores. The 40 GB Freecom FHD-1 came out on top on again, with a transfer rate of 33 MBps, followed closely by the Iomega 120 GB hard disk at 32 MBps. The sequential read rate is useful in determining the performance of a storage device, while working with very

large amounts of sequenced and ordered data such as a large DivX video file.

HDTach 2.7: In the tests conducted with HDTach, the 80 GB FHD-1 was the quickest. It had the lowest random access time of 13.8 milliseconds. Next came the lower capacity 40



Decision Maker

You routinely copy and transfer	You require	We recommend	Price
Dozens of PowerPoint presentations and Excel spreadsheets	A thumb drive with a 128 MB capacity and above.	Easydisk 128 MB, Transcend TS512MJFLASHA	Rs 3,500 to Rs 9,000
Large amounts of uncompressed video files, images, and the likes	A fast USB 2.0 drive with storage space of 40 GB and above	Iomega 120 GB HDD, Freecom FHD-1 40 GB	Rs 13,500 to Rs 15,000